

Original Investigation

Blood Culture Time to Positivity in Febrile Infants With Bacteremia

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IMPORTANCE Blood cultures are often obtained as part of the evaluation of infants with fever and these infants are typically observed until their cultures are determined to have no growth. However, the time to positivity of blood culture results in this population is not known.

OBJECTIVE To determine the time to positivity of blood culture results in febrile infants admitted to a general inpatient unit.

DESIGN, SETTING, AND PARTICIPANTS Multicenter, retrospective, cross-sectional evaluation of blood culture time to positivity. Data were collected by community and academic hospital systems associated with the Pediatric Research in Inpatient Settings Network. The study included febrile infants 90 days of age or younger with bacteremia and without surgical histories outside of an intensive care unit.

EXPOSURES Blood culture growing pathogenic bacteria.

MAIN OUTCOMES AND MEASURES Time to positivity and proportion of positive blood culture results that become positive more than 24 hours after placement in the analyzer.

RESULTS A total of 392 pathogenic blood cultures were included from 17 hospital systems across the United States. The mean (SD) time to positivity was 15.41 (8.30) hours. By 24 hours, 91% (95% CI, 88-93) had turned positive. By 36 and 48 hours, 96% (95% CI, 95-98) and 99% (95% CI, 97-100) had become positive, respectively.

CONCLUSIONS AND RELEVANCE Most pathogens in febrile, bacteremic infants 90 days of age or younger hospitalized on a general inpatient unit will be identified within 24 hours of collection. These data suggest that inpatient observation of febrile infants for more than 24 hours may be unnecessary in most infants.

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Blood cultures are routinely performed in the evaluation of infants with fever. These infants are typically hospitalized for 36 to 48 hours or longer to receive antibiotics while physicians await blood culture results.^{1,2}

Few data support a standard observational period for infants. Several prior studies have demonstrated that a substantial proportion of positive blood culture results may become positive after 24 or 36 hours,³⁻⁵ with some suggesting inpatient observation periods as long as 96 hours.^{6,7} However, these studies included infants admitted to intensive care units (ICUs) who are at higher risk for serious bacterial infection than the typical febrile infant and for pathogens that may have longer incubation times (eg, *Candida* species [spp]).³ Two studies argued

for a shorter period of observation; however, neither targeted the febrile infant 90 days old or younger.^{8,9} One large hospital system implemented discharge criteria of 24 to 36 hours without adverse consequences.¹⁰ Commonly used care guidelines¹¹ and textbooks^{2,12} disagree on the necessary period of observation and none are firm in their conclusions, reflecting the wide variation in practice in the management of infants with fever.

While there are several published guidelines for the treatment of febrile infants up to 90 days of age,^{13,14} we believe that the inconclusive nature of the current literature regarding the management of the febrile, but otherwise healthy, infant admitted to the general inpatient unit has hindered a standardized approach to the length of inpatient observation. Further-

more, the rarity of bacteremia in this population, estimated at 0.9% to 2%,^{10,15} and the geographical limitations of single-region microbiologic data¹⁶ necessitate a national approach. We present a large, geographically diverse examination of the time to positivity (TTP) of blood culture results in febrile, but otherwise healthy, infants 90 days of age or younger. Specifically, our primary aim was to determine the proportion of positive blood culture results that become positive more than 24 hours of incubation. As a secondary aim, we attempted to provide information regarding the epidemiology of bacteremia in this population and predictors of TTP in the most common bacterial species.

Methods

This multicenter, retrospective, cross-sectional evaluation of TTP of blood culture results included febrile infants 90 days of age and younger with bacteremia outside of the ICU. In total, 17 hospital systems affiliated with the Pediatric Research in Inpatient Settings Network across all major regions of the United States participated: Children's Hospitals and Clinics of Minnesota, Minneapolis; Children's Medical Center Dallas, Dallas, Texas; Cincinnati Children's Hospital Medical Center, Cincinnati, Ohio; Children's Hospital Colorado, Aurora; The Children's Hospital of The King's Daughters, Norfolk, Virginia; Children's Hospital Los Angeles, Los Angeles, California; Children's Hospital of Illinois, Peoria; Nemours/A.I. DuPont Hospital for Children, Wilmington, Delaware; Children's National Medical Center, Washington, DC; Albany Medical Center, Albany, New York; Tufts University School of Medicine, Boston, Massachusetts; University of Rochester Medical Center, Rochester, New York; Packard Children's Hospital at Stanford, Palo Alto, California; St Joseph's Mercy Hospital, Ann Arbor, Michigan; State University of New York Upstate, Syracuse; University of Iowa Children's Hospital, Iowa City; and Virginia Commonwealth University School of Medicine, Richmond.

Study dates varied by site secondary to the availability of data and feasibility of collection. However, every site collected at least 2 consecutive years of data ending January 1, 2013; the median duration of data collection was 4 years (range, 2-6 years). All sites used a BACTEC automated blood culture detection system throughout the collection period. Institutional review board approval with waiver of informed consent was obtained at each participating site.

Study Design

Each site obtained a data set from their microbiology laboratory of all positive blood culture results from infants 90 days of age and younger drawn in a non-ICU setting. A preliminary review of each record was performed by individual site investigators to determine eligibility. Cultures were included if the results were positive for bacteria, obtained from an infant 90 days old or younger with a temperature of 38.0°C or greater recorded on presentation or reported by caregiver, analyzed using a fully automated detection system, and treated as a pathogen by the medical team (this did not include empirical therapy and was specifically defined as prescribing a full course of antibiotics intended to treat the bacteria identified by blood

culture). Cultures drawn in any manner other than peripheral venipuncture in an ICU from patients admitted to the ICU within 5 hours after culture; from patients with central catheters or histories of intraabdominal, intracranial, or intrathoracic surgical procedures; or drawn from a hospital or clinic outside the system of any participating site were excluded.

For cultures meeting study criteria, site investigators collected patient age, sex, the location of the patient where the culture was drawn, antibiotics that were received within 24 hours preceding the culture, highest recorded temperature within 1 hour before or after obtaining the blood culture, blood culture TTP in minutes (calculated as the time at which the culture was placed in the detection system subtracted from the time the machine alarmed as positive), and the bacterial species identified. Each infant was stratified as low risk or nonlow risk based on a modification of the Rochester criteria used previously.^{10,14} As has been done previously, culture results from infants not meeting all of the low-risk criteria, even if all classifying data were unavailable, were placed in the nonlow-risk group.¹⁴

Data Analysis

Data on bacterial species were previously reported for 6 of the 17 participating sites; data on TTP were not previously reported for any site.¹⁶ Time to positivity was compared between groups using the Wilcoxon rank sum test. Because our study criteria allowed the inclusion of common contaminants if the physician team prescribed a full treatment course of antibiotics for the specific bacterium, we performed a comparison of TTP between all included cultures and all cultures after the removal of common contaminants (viridans group streptococci, coagulase-negative *Staphylococcus*, *Micrococcus* spp, *Corynebacterium* spp, and *Bacillus* spp).¹⁷ A multivariate analysis using a generalized linear model was performed to assess the impact of age, sex, prior antibiotics, and fever on TTP of the most common pathogenic bacterial species. Owing to the potential of type I error, the significance level, adjusted using Bonferroni correction, was set at .017 for this multivariate analysis.

To assess variability between sites, χ^2 analysis was used to compare the prevalence of 3 categories of bacteria: *Escherichia coli*, group B *Streptococcus* (GBS), and all other bacterial species. Bacterial prevalence was compared between the 8 highest-enrolling sites, with the remaining sites pooled into 1 group for comparison. A univariate analysis accompanied by a generalized linear model was performed to assess the variability in TTP between species and between study sites.

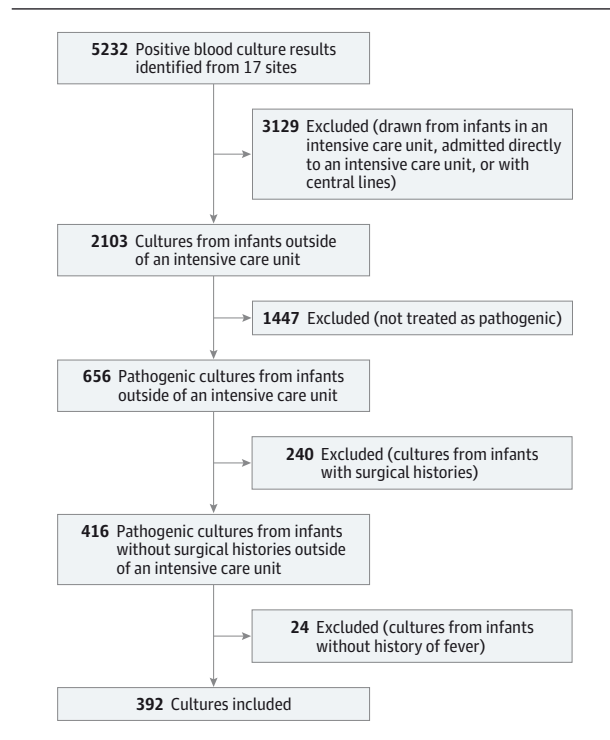
Data analysis was performed by 2 statisticians and the lead author (J.R., C.A., and E.A.B.). Statistical significance was set at a $P < .05$, unless otherwise noted, and all confidence intervals are reported at 95%. The statistical program used was SPSS version 21.

Results

Time to Positivity

A total of 5232 positive blood culture results were identified via initial search from the 17 systems. After exclusion criteria

Figure 1. Flow Diagram of Included and Excluded Infants and Blood Cultures



were applied, 392 pathogenic cultures remained for analysis (Figure 1). The mean (SD) TTP was 15.41 (8.30) hours and the median TTP was 13.00 hours (range, 5.22-68.30 hours). When compared with infants aged 61 to 90 days, there was a significantly shorter TTP in cultures drawn from infants 30 days and younger and 31 to 60 days old (Table 1). Cultures from infants with a temperature 38.0°C or greater within an hour of culture had a shorter TTP than those who did not (Table 1). Time to positivity was not significantly different in infants of different sexes or Rochester criteria risk stratification. When the 39 common contaminants (32 viridans group streptococci, 5 coagulase-negative *Staphylococcus*, 1 *Micrococcus* spp, and 1 *Bacillus* spp) were excluded, the TTP of the remaining cultures (mean [SD], 14.74 [7.18] hours) was not significantly different from the overall TTP ($P = .18$). Only 6 infants (2%) received antibiotics prior to obtaining the blood culture; therefore, this variable was not assessed in our model.

By 24 hours, 355 of the 392 blood culture results (91%; 95% CI, 88-93) were positive. By 36 and 48 hours, 378 (96%; 95% CI, 95-98) and 386 (99%; 95% CI, 97-100) had become positive, respectively (Figure 2). Of the bacteria growing after 24 hours, 30% (11 of 37) were *Escherichia coli* and 24% (9 of 37) were common contaminants (5 viridans group streptococci, 3 coagulase-negative *Staphylococcus*, and 1 *Micrococcus* spp). The median TTP was not statistically different in bacteria growing after 24 hours when common contaminants were included and removed (median, 31.9 vs 31.9 hours; $P = .89$).

A multivariate analysis was performed on the 4 most prevalent species known to be definite pathogens in culture (Table 2). For infants with *E coli* bacteremia, cultures drawn from those

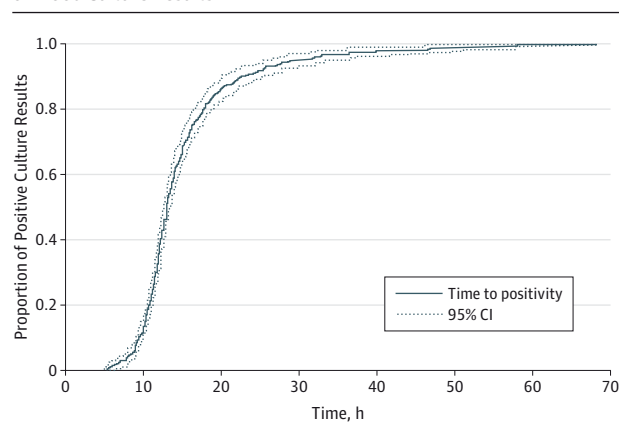
Table 1. Clinical Characteristics and Time to Positivity

Description	No. (%)	Time to Positivity, Mean (SD), h	P Value
Total No.	392 (100)	15.4 (8.3)	
Sex			
Male	201 (51)	15.4 (7.7)	.90
Female	191 (49)	15.5 (8.8)	
Age at blood drawn, d			
0-30	155 (40)	13.9 (4.9)	<.001
31-60	154 (39)	15.6 (9.0)	
61-90	83 (21)	17.9 (10.8)	
Highest temperature recorded within 1 h, C ^a			
<38.0	107 (27)	17.0 (9.9)	.006
≥38.0	210 (54)	14.9 (7.8)	
Infant risk stratification			
Low	99 (25)	15.7 (9.1)	.56
Nonlow	293 (75)	15.3 (8.0)	
Site			
1	87 (22)	14.9 (5.4)	1 [Reference]
2	53 (14)	17.0 (8.6)	
3	44 (11)	19.3 (13.5)	<.001
4	36 (9)	13.7 (4.3)	
5	31 (8)	14.1 (8.9)	.45
6	27 (7)	14.5 (6.8)	
7	23 (6)	12.9 (6.4)	.13
8	17 (4)	16.6 (11.8)	
9	14 (4)	14.3 (3.2)	.69
10-17	60 (15)	14.9 (7.9)	

Abbreviation: NA, not applicable.

^a Seventy-five (19%) had no temperature recorded within 1 hour of the blood culture.

Figure 2. Kaplan-Meier Curve Representing the Time to Positivity of Blood Culture Results



in the 0- to 30-day age range were more likely to have shorter TTP than those in the 61- to 90-day age range (mean, 13.3 vs 16.1 hours; $P = .009$). Cultures from infants with *Staphylococcus aureus* bacteremia in the 31- to 60-day age range took longer to grow than cultures from infants in the 61- to 90-day age

Table 2. Multivariate Analysis of Demographics and Mean Time to Positivity Within Common Species^a

Demographic	<i>E coli</i>			GBS			<i>S aureus</i>			<i>S pneumoniae</i>		
	No.	Mean (SD)	P Value	No.	Mean (SD)	P Value	No.	Mean (SD)	P Value	No.	Mean (SD)	P Value
Sex												
Female	78	15.3 (7.0)	.27	47	11.1 (3.7)	.31	11	17.8 (5.8)	.55	10	16.4 (7.1)	.30
Male	81	13.9 (3.9)		40	11.2 (5.6)		12	20.7 (6.5)		8	15.8 (5.8)	
Age, d												
0-30	68	13.3 (3.3)	.009	34	11.2 (5.6)	.98	8	17.2 (3.2)	.72	2	14.4 (3.0)	.44
31-60	58	15.2 (6.9)	.65	41	11.2 (5.6)	.54	11	21.1 (8.2)	.01	5	18.7 (10.0)	.11
61-90	33	16.1 (6.7)	1 [Reference]	12	10.6 (2.7)	1 [Reference]	4	18.7 (2.9)	1 [Reference]	11	15.2 (4.8)	1 [Reference]
Risk												
Low	19	13.4 (4.3)	.25	34	10.5 (3.2)	.01	7	19.0 (2.2)	.23	3	22.6 (11.7)	.04
Nonlow	140	14.7 (5.9)		53	11.5 (5.4)		16	19.5 (7.4)		15	14.8 (4.3)	
Fever, °C												
<38.0	36	14.8 (4.6)	.68	21	10.5 (2.9)	.08	8	21.1 (8.1)	.08	3	20.0 (7.6)	.06
≥38.0	93	15.0 (6.6)		54	10.9 (4.0)		13	18.8 (5.2)		11	15.5 (7.0)	

Abbreviation: GBS, group B *Streptococcus*.^a Bonferroni correction significance set at 0.017.

Table 3. Epidemiology of Bacteremia and Generalized Linear Model Comparing Median Time to Positivity Between Species

Organism	No. (%)	Time to Positivity, Median (Range), h	P Value
<i>E coli</i>	159 (41)	13.0 (7.8-53.3)	1 [Reference]
GBS	87 (22)	10.5 (5.2-39.5)	<.001
Viridans group streptococci ^a	32 (8)	17.2 (10.3-57.2)	<.001
<i>S aureus</i>	23 (6)	18.5 (12.4-40.1)	<.001
<i>S pneumoniae</i>	18 (5)	14.3 (11.2-36.1)	.19
<i>Enterococcus</i> spp	14 (4)	15.3 (9.8-25.7)	.46
<i>Klebsiella</i> spp	10 (3)	12.0 (8.9-15.0)	.10
<i>Enterobacter</i> spp	8 (2)	13.0 (9.2-19.0)	.38
<i>Salmonella</i> spp	6 (2)	15.1 (12.3-22.0)	.44
<i>S pyogenes</i>	5 (1)	13.6 (9.1-37.3)	.10
Other γ-hemolytic <i>Streptococcus</i>	5 (1)	14.1 (5.6-14.8)	.20
CoNS ^a	5 (1)	27.2 (15.0-68.3)	<.001
<i>Moraxella</i> spp	4 (1)	39.8 (22.7-56.0)	<.001
<i>Neisseria</i> spp	3 (1)	23.5 (17.6-27.2)	.003
<i>Serratia</i> spp	2 (0)	17.0 (16.2-17.8)	.46
<i>Pseudomonas</i> spp	2 (0)	29.0 (11.9-46.0)	<.001
<i>Micrococcus</i> spp ^a	1 (0)	58.8 (58.8-58.8)	<.001
Other	8 (2)	19.9 (9.4-58.2)	<.001

Abbreviations: GBS, group B *Streptococcus*; CoNS, coagulase-negative *Staphylococcus*; spp, species.^a Considered common contaminants.

range (mean, 21.1 vs 18.7 hours, respectively; $P = .01$). Time to positivity did not differ significantly in infants with temperatures 38.0°C or greater within 1 hour of obtaining the culture and those with temperatures of less than 38.0°C.

Epidemiology and Variability

E coli was the most prevalent species (159 of 392, 41%), followed by GBS (87 of 392, 22%), viridans group streptococci (32 of 392, 8%), *S aureus* (23 of 392, 6%), and *Streptococcus pneumoniae* (18 of 392, 5%). *E coli* was also the most common species identified in infants 30 days or younger (68 of 155, 44%), followed by GBS (34 of 155, 22%). Group B *Streptococcus* had

the shortest median TTP among all species identified (10.5 hours; range, 5.2-39.6 hours). Our linear model suggested significant variability in TTP between bacterial species, as the TTP of the reference bacteria (*E coli*) was significantly different from several other species including GBS, *S aureus*, and viridans group streptococci (Table 3).

A generalized linear model was used to compare bacterial species between sites for variation in TTP. Data from the largest site were used as a reference. There was statistically significant TTP variation of *E coli* in 5 of the 17 sites, of GBS in 2 sites, and of *S aureus* in 1 site. *S pneumoniae* showed no variation in TTP between sites (Table 1).

Discussion

While inpatient evaluation of an otherwise healthy, febrile infant is a common clinical scenario encountered by physicians and discharge is often based on negative culture results, there is a lack of consensus on the time of inpatient observation while awaiting culture results. Furthermore, the rarity of bacteremia in this population¹⁰ and the variability of bacterial epidemiology between geographic regions in the United States¹⁶ warrants a national approach to this question.

In our large geographically diverse investigation, we estimated that 91%, 96%, and 99% of blood cultures from this population will turn positive by 24, 36, and 48 hours, respectively. Assuming an incidence of 0.9% to 2% for bacteremia in this population,^{10,15} our data suggest an observation period beyond 24 hours would capture 1 additional bacteremic infant for every 556 to 1235 febrile infants evaluated. Similarly, observation for more than 36 hours and more than 48 hours would identify 1 bacteremic infant for every 1250 to 2778 and 5000 to 11111 infants, respectively.

A standard inpatient observation time of 48 to 72 hours for blood cultures in infants with fever was initially established when cultures were manually observed for bacterial growth at infrequent intervals.^{18,19} Automated, continuous-monitoring blood culture systems have since been instituted in most laboratories and are able to detect bacterial growth significantly sooner than manual methods.^{20,21} In addition to advances in blood culture monitoring, the changing epidemiology of pathogens causing bacteremia has resulted in evolving approaches to the evaluation and management of febrile infants.^{10,16,22} There has also been a decrease in the reported rates of occult bacteremia since vaccinations for *Haemophilus influenzae* type B and *S pneumoniae* became universally available in the United States, as well as a decrease in early-onset sepsis caused by GBS since the implementation of widespread intrapartum antibiotic prophylaxis.²³⁻²⁵

Recent studies have reported a decrease in the TTP for blood culture results in pediatric patients; however, these studies were small single-center examinations or focused on populations very different from the classic febrile infant admitted to a general inpatient unit for observation.^{3,8,9,23,26-28}

Our data suggest that 24 hours (as opposed to the generally accepted 48-hour inpatient observation period) is adequate to detect most clinically significant bacteremia and by using this cutoff, we can minimize the number of nights spent in the hospital for these infants and caregivers. We were unable to examine the TTP of urine and cerebrospinal fluid cultures, typically followed as part of the febrile infant evaluation; however, analyses of urine and cerebrospinal fluid samples (eg, white blood cell count and Gram stain) can be performed within hours of presentation and have high sensitivity for urinary tract infection and meningitis, respectively.^{15,29,30}

While there remains an established risk for serious bacterial infection in young infants with fever, the expanding body of research leads the practitioner to continually weigh the benefits of treatment and the risks posed by health care interven-

tions. Hospitalized patients are at risk for iatrogenic complications such as intravenous infiltration, hospital-acquired infections, and adverse medication effects.^{31,32} Parents also experience significant stress when a child is hospitalized in terms of both concern for the child's health and time spent away from other daily responsibilities.³³ Additionally, hospital charges for the evaluation of low-risk febrile infants are estimated at \$6613 per infant (adjusted to 2013 dollars).^{31,32} Finally, recent data suggest the incidence of hospital-acquired infection on a general pediatric unit is 1 per 1000 patient-days³⁴; therefore, it can be estimated that for every case of bacteremia identified at more than 24 hours in infants who remain hospitalized, 1 nosocomial infection will also occur.

As has been done previously,¹⁶ we defined pathogenic bacteria as those that were treated with a full course of antibiotics rather than simply excluding bacteria commonly considered to be contaminants. While this method did result in the inclusion of several cultures that are often considered contaminants, it did not significantly alter the overall TTP and allowed us to exclude many cases in which contamination may have otherwise been equivocal (eg, *Acinetobacter* spp).

Our study had several limitations. First, we were unable to account for the time between collection of the blood culture and placement in the machine. Blood is typically drawn directly into culture media; therefore, our data may underestimate the total time needed for bacteria to grow. While the hospitals in our study have different blood draw protocols, it is generally assumed that cultures take less than 30 minutes from collection to placement in the automated detection system. Second, criteria for ICU admission vary by institution and this may have introduced some unidentified heterogeneity in our population. Third, there was some variability in TTP of a few specific bacterial species at a minority of sites. While it is possible that there are specific bacterial attributes that differ geographically or that the differing inclusion dates between some sites introduced variation in TTP, another plausible explanation is that we were unable to control for the amount of blood inoculated, which does impact TTP.³⁵ Fourth, we were unable to identify clinical predictors for a culture result taking more than 24 hours to become positive and therefore cannot suggest likely outcomes for these infants should they be discharged prior to the culture result becoming positive. Furthermore, because we did not collect data regarding negative culture results, we cannot provide information regarding overall risk for serious bacterial infection to our population or by subgroup. Finally, our subgroup analyses included testing for multiple variables and while we did use Bonferroni correction for multiple testing, there was the possibility of type I error in those analyses.

Conclusions

Our study found that most pathogens in febrile bacteremic infants 90 days of age and younger will be identified within 24 hours of collection. These data could support a national standardized approach to the duration of inpatient observation of the febrile infant.

ARTICLE INFORMATION

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Study concept and design: Biondi, Mischler.

Acquisition, analysis, or interpretation of data: All authors.

Drafting of the manuscript: Biondi, Mischler, Jerardi, French, Evans, Lee, Chen.

Critical revision of the manuscript for important intellectual content: Biondi, Mischler, Jerardi, Statile, Evans, Asche, Ren, Shah.

Statistical analysis: Biondi, Mischler, Asche, Ren.

Obtained funding: Biondi.

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Conflict of Interest Disclosures: None reported.

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Disclaimer: The authors have not published substantially similar papers from this study, although some of the pilot data regarding epidemiology were published last year. The data on time to positivity, the focus of this article, have not been previously published. Dr Shah is the *JAMA Pediatrics* Clinical Challenge Section Editor but was not involved in the review process or the acceptance of the manuscript.

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REFERENCES

- Biondi E, Murzycki J, Ralston S, Gigliotti F. Fever and bacteremia. *Pediatr Rev*. 2013;34(3):134-136.
- Remington JS. *Infectious Diseases of the Fetus and Newborn Infant*. 7th ed. Philadelphia, PA: Saunders/Elsevier; 2011.
- Garcia-Prats JA, Cooper TR, Schneider VF, Stager CE, Hansen TN. Rapid detection of microorganisms in blood cultures of newborn infants utilizing an automated blood culture system. *Pediatrics*. 2000;105(3, pt 1):523-527.
- Guerti K, Devos H, Ieven MM, Mahieu LM. Time to positivity of neonatal blood cultures: fast and furious? *J Med Microbiol*. 2011;60(Pt 4):446-453.
- Kumar Y, Qunibi M, Neal TJ, Yoxall CW. Time to positivity of neonatal blood cultures. *Arch Dis Child Fetal Neonatal Ed*. 2001;85(3):F182-F186.
- Randhawa VS, Sherwal BL, Mehta G. Incubation period for culture positivity to detect septicaemia in neonates. *Indian J Med Microbiol*. 2006;24(3):237-238, author reply 238.
- Kara A, Kanra G, Cengiz AB, Apış M, Gür D. Pediatric blood culture: time to positivity. *Türk J Pediatr*. 2004;46(3):251-255.
- McGowan KL, Foster JA, Coffin SE. Outpatient pediatric blood cultures: time to positivity. *Pediatrics*. 2000;106(2, pt 1):251-255.
- Kaplan RL, Harper MB, Baskin MN, Macone AB, Mandl KD. Time to detection of positive cultures in 28- to 90-day-old febrile infants. *Pediatrics*. 2000;106(6):74.
- Byington CL, Reynolds CC, Korgenski K, et al. Costs and infant outcomes after implementation of a care process model for febrile infants. *Pediatrics*. 2012;130(1):e16-e24.
- Team FUS, Cincinnati Children's Hospital Medical Center. Evidence-based clinical care guideline for fever of uncertain source in infants 60 days of age or less. Guideline 02, pages 1-14, October 2010. <http://www.cincinnatichildrens.org/svc/alpha/h/health-policy/ev-based/default.htm>. Accessed April 17, 2014.
- Kliegman R, Nelson WE. *Nelson Textbook of Pediatrics*. 19th ed. Philadelphia, PA: Elsevier/Saunders; 2011.
- Bachur RG, Harper MB. Predictive model for serious bacterial infections among infants younger than 3 months of age. *Pediatrics*. 2001;108(2):311-316.
- Jaskiewicz JA, McCarthy CA, Richardson AC, et al; Febrile Infant Collaborative Study Group. Febrile infants at low risk for serious bacterial infection: an appraisal of the Rochester criteria and implications for management. *Pediatrics*. 1994;94(3):390-396.
- Herr SM, Wald ER, Pitetti RD, Choi SS. Enhanced urinalysis improves identification of febrile infants ages 60 days and younger at low risk for serious bacterial illness. *Pediatrics*. 2001;108(4):866-871.
- Biondi E, Evans R, Mischler M, et al. Epidemiology of bacteremia in febrile infants in the United States. *Pediatrics*. 2013;132(6):990-996.
- Bekeris LG, Tworek JA, Walsh MK, Valenstein PN. Trends in blood culture contamination: a College of American Pathologists Q-Tracks study of 356 institutions. *Arch Pathol Lab Med*. 2005;129(10):1222-1225.
- Pichichero ME, Todd JK. Detection of neonatal bacteremia. *J Pediatr*. 1979;94(6):958-960.
- Rowley AH, Wald ER. Incubation period necessary to detect bacteremia in neonates. *Pediatr Infect Dis*. 1986;5(5):590-591.
- Krisher KK, Whyburn DR, Koepnick FE. Comparison of the BacT/Alert pediatric blood culture system, Pedi-BacT, with conventional culture using the 20-milliliter Becton-Dickinson supplemented peptone broth tube. *J Clin Microbiol*. 1993;31(4):793-797.
- Magadia RR, Weinstein MP. Laboratory diagnosis of bacteremia and fungemia. *Infect Dis Clin North Am*. 2001;15(4):1009-1024.
- Greenhow TL, Hung YY, Herz AM. Changing epidemiology of bacteremia in infants aged 1 week to 3 months. *Pediatrics*. 2012;129(3):e590-e596.
- Alpern ER, Alessandrini EA, Bell LM, Shaw KN, McGowan KL. Occult bacteremia from a pediatric emergency department: current prevalence, time to detection, and outcome. *Pediatrics*. 2000;106(3):505-511.
- Centers for Disease Control and Prevention (CDC). Progress toward elimination of Haemophilus influenzae type b invasive disease among infants and children: United States, 1998-2000. *MMWR Morb Mortal Wkly Rep*. 2002;51(11):234-237.
- Schrag S, Gorwitz R, Fultz-Butts K, Schuchat A. Prevention of perinatal group B streptococcal disease: revised guidelines from CDC. *MMWR Recomm Rep*. 2002;51(rr-11):1-22.
- Hurst MK, Yoder BA. Detection of bacteremia in young infants: is 48 hours adequate? *Pediatr Infect Dis J*. 1995;14(8):711-713.
- Evans RC, Fine BR. Time to detection of bacterial cultures in infants aged 0 to 90 days. *Hosp Pediatr*. 2013;3(2):97-102.
- Friedman J, Matlow A. Time to identification of positive bacterial cultures in infants under three months of age hospitalized to rule out sepsis. *Paediatr Child Health*. 1999;4(5):331-334.
- Shaw KN, McGowan KL, Gorelick MH, Schwartz JS. Screening for urinary tract infection in infants in the emergency department: which test is best? *Pediatrics*. 1998;101(6):1.
- Dunbar SA, Eason RA, Musher DM, Clarridge JE III. Microscopic examination and broth culture of cerebrospinal fluid in diagnosis of meningitis. *J Clin Microbiol*. 1998;36(6):1617-1620.
- DeAngelis C, Joffe A, Wilson M, Willis E. Iatrogenic risks and financial costs of hospitalizing febrile infants. *AJDC*. 1983;137(12):1146-1149.
- Condra CS, Parbhu B, Lorenz D, Herr SM. Charges and complications associated with the medical evaluation of febrile young infants. *Pediatr Emerg Care*. 2010;26(3):186-191.
- Nizam M, Norzila MZ. Stress among parents with acutely ill children. *Med J Malaysia*. 2001;56(4):428-434.
- Yin J, Schweizer ML, Herwaldt LA, Pottinger JM, Perencevich EN. Benefits of universal gloving on hospital-acquired infections in acute care pediatric units. *Pediatrics*. 2013;131(5):e1515-e1520.
- Isaacman DJ, Karasic RB, Reynolds EA, Kost SI. Effect of number of blood cultures and volume of blood on detection of bacteremia in children. *J Pediatr*. 1996;128(2):190-195.